Listen Before You Speak · AL · Practice

24 written + 6 = 30 marks

Name: _____ Due: _____

9618 · Media, network hardware, Ethernet and ports. Q1–Q2 are real anchors; Q3–Q7 are ours.

1. November 2022 Paper 11 Q8 · © UCLES.

A Local Area Network (LAN) uses a bus topology.

Describe how Carrier Sense Multiple Access / Collision Detection (CSMA / CD) is used in a bus network. [4]

2. November 2021 Paper 11 Q8(b)(c) · © UCLES.

A school is setting up a network within one of its buildings.

One classroom in the building has 30 computers. The computers need to be connected to the network. Each computer has a network interface card (NIC).

(b) Identify two possible devices that can be used to physically connect the 30 computers to the rest of the network. [2]

2. School network — continued

The school has several laptops. Each laptop has a Wireless Network Interface Card (WNIC).

(c) Describe the functions of a Wireless Network Interface Card. [4]

3. Ours: a warehouse yard.

(a) Recommend fibre rather than copper for a link past heavy electrical machinery. Give two comparative reasons. [2]

(b) Forklift scanners move around the yard. Give one benefit and one drawback of wireless rather than a wired connection for these scanners. [2]

4. Ours: choose the hardware.

Name the device for each role: (i) forward LAN frames using a table of MAC addresses; (ii) connect wireless clients to a wired LAN; (iii) regenerate a weakened signal. [3]

5. Ours: the router at the school boundary.

Describe how the router forwards a packet addressed to a host on a different network. Give three steps beyond “connects to the internet”. [3]

6. Ours: the presentation computer.

(a) A display has built-in speakers. Give two advantages of HDMI over VGA for this connection. [2]

(b) A wired keyboard connects by USB. State the transmission method USB uses and one other useful feature of USB for this peripheral. [2]

7. The Wire — pen first, then code.

File: challenge_U14b_TheWire.py

A frame needs four consecutive send slots. A and B each receive a frame in slot 1; C receives one in slot 10. Decisions use the previous slot's busy/collision flags. After a collision, the next slot aborts/jams; A draws a wait of 1 whole slot, B draws 0. A waiting station counts down before listening again. Trace slots 1–16 before running **python3 challenge_U14b_TheWire.py trace**.

Trace symbols: S send, L listen, X collide, J jam, W wait, – idle.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
A																
B																
C																

(a) Pen first. Sketch a slot trace for A, B and C. State the collision slot and the slot when each frame completes. Trace is self/teacher checked. [2]

(b) Then code. Implement Station.tick: listen and send a complete frame [1]; abort without losing the queued frame and retry [1]; random doubling back-off and successful network tests [1]. Run the file's judge: its six checks support these three marks. [3]

(c) Explain why using the same fixed delay after every collision could cause repeated collisions. [1]

Above and beyond · unmarked. Modern switched full-duplex Ethernet does not use this collision procedure. Explain what changed about who shares each transmission direction.

Additional space — label the question.